Alaittin Kirtisoglu

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About

My research interests lie, broadly speaking, in discrete mathematics and operations research, especially graph theory, network optimization and related models, algorithms, and interdisciplinary applications in organ allocation, healthcare networks and political geometry.

Education

Ph.D. in Applied Mathematics, Illinois Institute of Technology, Chicago *Aug 2021 -Aug 2026*

- Dissertation: Contributions to hierarchical facility location and districting problems
- Supervisor: Hemanshu Kaul

M.Sc. in Mathematics, Hacettepe University, Turkey *Feb 2019 - Jun 2021*

- Dissertation: Variations of star coloring on graphs
- Supervisor: Lale Özkahya, GPA: 3.93/4.00

B.Sc. in Mathematics, Mustafa Kemal University, Turkey *Sep 2014 - Jun 2018*

- GPA: 3.45/4.00 (1st place)
- Exchange student at Lublin University of Technology, Poland (Feb–July 2016)

Experience

Research Scientist - NYU Grossman School of Medicine, Department of Surgery *Nov 2026–Present*

- Transplant-focused operations research; organ allocation optimization, machine learning and statistical inference, programming discrete event simulations, grant development for NIH-funded projects.

Graduate Teaching Assistant - Illinois Institute of Technology *Aug 2021–May 2026*

- *Undergraduate level*: calculus, linear algebra, differential equations, complex analysis, discrete mathematics, and computational mathematics.
- *Graduate level*: linear optimization, regression, statistical learning, time series, mathematical modeling, and computational algebraic geometry.

Research Intern - Fermi National Accelerator Laboratory *15 May - 22 July 2022*

- I was a part of the Exa.TrkX project, working on hierarchical graph neural networks for particle trajectory reconstruction in 3D detector data

Graduate Teaching Assistant - TED University, Turkey *Sep 2019 - Aug 2021*

- Undergraduate courses in calculus, linear algebra, and differential equations.

Mentorship

Research mentor *current*

- Advising a senior undergraduate student on Paper (5) in the list of Papers.

Mentor - Directed Reading Program *2022–2026*

- Advised six undergraduate mathematics students on nine-week summer research projects in optimization, machine learning, graph theory, and statistical learning, each ending with a written report and a symposium presentation.

Fellowship

- SoReMo (Socially Responsible Modeling) Research Fellowship 2023

  Awarded for socially equitable approaches in computation and modeling, for the FalCom paper.

- European Union Marie Skłodowska-Curie Actions Postdoctoral Fellowship (Under review) 2026
  Submitted a proposal on *"A joint sampling approach for districting and routing in an Istanbul Earthquake Scenario"*.
  Host: Duygu Tas Kuten, Sabanci University, Istanbul, Turkey.

Papers

Published

1. ***Coloring of Graphs Avoiding Bicolored Paths of a Fixed Length***, (with Lale Özkahya)
   Graphs and Combinatorics, 40 (2024), Art. 11.

   We study proper vertex colorings of graphs that avoid bicolored paths of a fixed length. We introduce the P_k-chromatic number $s_k(G)$, the minimum number of colors needed for such a coloring. For any graph with maximum degree $d \geq 2$ and $k \geq 4$, we establish that $s_k(G) = O\big(d^{(k-1)/(k-2)}\big)$. We also determine exact values of s_k for products of cycles and paths when $k = 5$ and $k = 6$. The problem generalizes classical star coloring and acyclic coloring and connects chromatic theory with Ramsey-type structural questions.

Submitted

2. ***FalCom: A Sampling Method for Districting and Hierarchical Facility Location*** (with Hemanshu Kaul)
   Extended abstract presented at the 9th International Symposium on Combinatorial Optimization (ISCO).

   Districting partitions geographic areas into balanced, contiguous, and compact districts. Recent advances in political redistricting have demonstrated the potential of recombination-type Markov chain Monte Carlo (MCMC) methods for exploring the space of feasible districting plans and revealing structural alternatives. In this paper, we introduce *FalCom* (Facility Location Combination), a MCMC sampler for hierarchical districting with capacitated facility assignments. FalCom partitions a geographic region at two hierarchical levels, assigns a facility and a capacity to every district and balance districts according to the capacity constraints. We apply FalCom to the London Ambulance Service area including 66 real ambulance stations. The resulting ensemble characterizes the variation of response-area boundaries, station workloads and operational groupings under the sampler's law, and provides a distributional reference for assessing the existing service layout.

3. ***Fixed-Wing UAV Routing for Time-Dependent Information Collection with Speed and Loiter Decisions*** (with Melis Boran and Mustafa Kemal Tural). We study a routing problem in which a fixed-wing unmanned aerial vehicle (UAV) selects and orders targets with time windows to maximize information collection, which varies linearly with arrival time. The UAV chooses continuous speeds on flight legs and may loiter between targets, while a nonlinear propulsion model governs energy consumption. We prove that constant-speed flight on each leg is energy-optimal and formulate the problem as a mixed-integer nonlinear program with an exact mixed-integer second-order cone reformulation. We develop an iterated-local-search matheuristic that evaluates candidate routes through convex second-order cone programs. Experiments calibrated to realistic UAV missions show that the exact approach is limited to small or tightly windowed instances, whereas the matheuristic scales to hundreds of targets and matches or improves upon exact-solver solutions.

Preprint

4. ***Multi-Place Robot Rendezvous Problem with Campaign Time and Energy Consumption Restrictions*** (with Nazli Dolu Hasturk and Mustafa Kemal Tural).

   We study the energy-efficient multi-place robot rendezvous problem (EEMPR), in which a single tanker robot recharges a team of mobile worker robots by meeting each one at a rendezvous place, in a given meeting order. Robots choose their speeds and consume energy through a nonlinear function of speed, and worker batteries are bounded by a capacity and a minimum level. For a fixed meeting order we show that the problem is a second-order cone program (SOCP), which is solvable to optimality in polynomial time. We then search over meeting orders with 2-opt and 3-opt edge exchange heuristics and speed-up techniques of Travel Salesman Problem, by solvingone SOCP per candidate meeting order.

In progress

5. ***The Diffusion Walk: Sampling Graph Partitions with Kernel-Aware Local Refinements*** (with Onur Ongun).

   ReCom variants, the dominant MCMC methods for political redistricting, are computationally efficient but geometrically blind. We propose a hybrid sampler that complements ReCom with a kernel-aware local-refinement step, based on graph Lapacian formulated as a quadratic programming model.

6. ***Optimization-Based Decision Support for Equitable Expansion of Federally Qualified Health Centers in Chicago*** (with Kim Erwin and Hemanshu Kaul).

   We identify four critical gaps in the current granting system of Health Resources and Services Administration (HRSA) for Federally Qualified Health Centers (FQHCs): the absence of location-evaluation tools; an outdated Index of Medical Underservice (IMU) formula; a limited Medically Underserved Area (MUA) designation process; and unmapped referral networks between FQHCs and hospitals. We show how optimization and network analysis can address each. To the best of our knowledge, this is the first paper bringing solutions from operations research to the location decisions of FQHCs for policy-makers.

External service

- European Symposium on Algorithms (ESA) - Sub-reviewer 2026

Talks

1. Middle East Technical University, Industrial Engineering Departmental Seminars May 2026
2. International Symposium on Combinatorial Optimization (ISCO) May 2026
3. Illinois Institute of Technology, Discrete Math Seminars Apr 2026
4. European Combinatorial Conference (online) Aug 2021
5. SIAM Conference on Discrete Mathematics (online) July 2021
6. British Combinatorial Conference (online) June 2021

Software and decision tools

- FalCom dashboard - Interactive visualization tool for ensemble analysis and policy comparison for hierarchical capacitated districting and facility location.
- London Ambulance Service map - Interactive visualization of the hierarchical districting of the London area and supply-demand balance for the LAS application of FalCom.
- Chicago Healthcare Network dashboard - Decision-support tool for equitable Federally Qualified Health Center expansion in Chicago with optimization, transit-time-aware accessibility, and scenario comparisons.
- FalcomChain, FalcomPlot, FalcomTravel - FalCom family of open-source Python libraries for running a Falcom chain, plotting spatial inputs and outputs of Falcom chains, and travel-time-matrix computations.

Technical skills

Programming

- Python, R, Javascript, MATLAB, HTML

Optimization & modeling

- Gurobi, CPLEX, metaheuristics, flow and stochastic models, integer programming, Markov Chain Monte Carlo

Libraries for geospatial & scientific computing

- NetworkX, NumPy/SciPy, Pandas, Matplotlib, GeoPandas, OSMnx, Folium, ArcGIS, GerryChain

Machine learning

- PyTorch, TensorFlow, scikit-learn, graph neural networks, reinforcement learning, regression analysis